

■ SQL Revision Notes: Data Manipulation (INSERT, UPDATE, DELETE)

1. The Importance of DML

DML commands are used to modify the data stored in a database. Unlike SELECT, these commands change the data permanently.

Command	Purpose	Simple Analogy
INSERT	To add a new row (record) to a table.	Adding a new student's file to a school's cabinet.
UPDATE	To modify existing data in one or more rows.	Changing a student's address or phone number in their existing file.
DELETE	To remove one or more existing rows from a table.	Permanently removing a student's file from the cabinet.

2. INSERT: Adding New Data

The INSERT statement is used to put new rows into a table.

A. Inserting Values into ALL Columns

INSERT INTO TableName VALUES (Value1, Value2, Value3, ...); -- Example: Adding a new product
INSERT INTO Products VALUES (101, 'Widget X', 25.00, 'Electronics', 150);

B. Inserting Values into SPECIFIC Columns

INSERT INTO TableName (Column1, Column2, Column3) VALUES (ValueA, ValueB, ValueC); --
Example: Adding a customer who hasn't provided an email yet
INSERT INTO Customers (CustomerID, FirstName, LastName) VALUES (5001, 'Alice', 'Smith');

3. UPDATE: Changing Existing Data

The UPDATE statement is used to modify data in an existing table. The WHERE clause is critical here.

A. Basic Syntax

UPDATE TableName SET Column1 = NewValue1, Column2 = NewValue2, ... WHERE Condition;

B. Crucial Role of WHERE

With WHERE: The change is applied only to the rows that meet the condition.

Example: Give a 10% raise to only the managers. UPDATE Employees SET Salary = Salary * 1.10
WHERE JobTitle = 'Manager';

Without WHERE: The change is applied to ALL rows in the table! (Use with extreme caution!)

Example: Set the 'Status' column to 'Active' for everyone. UPDATE Orders SET Status = 'Active'; --
WARNING: Affects all orders!

4. DELETE: Removing Data

The DELETE statement is used to remove one or more rows from a table. The WHERE clause is equally critical here.

A. Basic Syntax

DELETE FROM TableName WHERE Condition;

B. Crucial Role of WHERE

With WHERE: The row(s) that meet the condition are removed.

Example: Remove only the products that are out of stock. DELETE FROM Products WHERE StockQuantity = 0;

Without WHERE: ALL rows are removed from the table (leaving the table structure intact). (Use with extreme caution!)

Example: Clear out the entire log history. DELETE FROM Logs; -- WARNING: Deletes all log entries!

5. Safety First: The DML Mantra

Whenever you write an UPDATE or DELETE statement, adopt this safety procedure:

■ Write the SELECT Query First:

Before running UPDATE or DELETE, write the equivalent SELECT statement with the exact same WHERE clause.

Example: Instead of running DELETE FROM Products WHERE Price > 100;, first run: SELECT * FROM Products WHERE Price > 100;

■ Verify the Results:

Check the result set of the SELECT query. Are these the exact rows you intend to modify or delete?

■ Execute the DML:

If the results are correct, execute the UPDATE or DELETE statement.

This process ensures you never accidentally alter or delete data you didn't intend to touch.